1) a)
2^{k+1}

1) b)
-3^{k+1}

1) c)
a^{k+1}

2) a)
$(1+a)^{k+1}$

2) b)
$(1-a)^{k+1}$

3) a)
$3!$

3) b)
$12!$

3) c)
$(k+1)!$

4)

Prove true for $n = 3$:

$2^3 = 8$ and $2 \times 3 = 6$ and since $8 \geq 6$ so true for $n = 3$

Assume true for $n = k$:

$2^k > 2k$

Prove true for $n = k + 1$:

$2^{k+1} = 2 \times 2^k$

So since $2^k > 2k$

$2^{k+1} > 2 \times 2k$

$2^{k+1} > 4k$

$2^{k+1} > 2k + 2k$

Since $k \geq 3$, we know that $2k \geq 6$

$2^{k+1} > 2k + 6$

$2^{k+1} > 2k + 2 + 4$

So $2^{k+1} > 2k + 2$

$2^{k+1} > 2(k + 1)$

which is $2^n > 2n$ with $n = k + 1$ substituted in

As the statement is true for $n = 3$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 3$

5)

Prove true for $n = 2$:

$4^2 = 16$ and $3 \times 2 + 7 = 13$ and since $16 > 13$ so true for $n = 2$

Assume true for $n = k$:

$4^k > 3k + 7$

Prove true for $n = k + 1$:

$4^{k+1} = 4 \times 4^k$

So since $4^k > 3k + 7$ then

$4 \times 4^k > 4(3k + 7)$

$4 \times 4^k > 12k + 28$

So $4^{k+1} > 12k + 28$

But as $k \geq 2$ then $12k + 28 > 3k + 10$

$4^{k+1} > 3k + 10$

$4^{k+1} > 3k + 3 + 7$

So it holds that $4^{k+1} > 3(k + 1) + 7$

which is $4^n > 3n + 7$ with $n = k + 1$ substituted in

As the statement is true for $n = 2$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 2$

6)

Prove true for $n = 5$:

$5^2 = 25$ and $2^5 = 32$ and since $25 < 32$ true for $n = 5$

Assume true for $n = k$:

$k^2 < 2^k$

Prove true for $n = k + 1$:

$2k^2 < 2 \times 2^k$

$2k^2 < 2^{k+1}$

$k^2 + k^2 < 2^{k+1}$

But since $k > 4$ then $k^2 > 4k$

$k^2 + 4k < 2^{k+1}$

$k^2 + 2k + 2k < 2^{k+1}$

But also, since $k > 4$ then $2k > 8$

$k^2 + 2k + 8 < 2^{k+1}$

$(k + 1)^2 + 7 < 2^{k+1}$

So it holds that $(k + 1)^2 < 2^{k+1}$

Which is $n^2 < 2^n$ with $n = k + 1$ substituted in

As the statement is true for $n = 5$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 5$

7) a)

$3^3 = 27$ and $3^3 = 27$ so it doesn't hold when $n = 3$

$a = 3$

7) b)

When $n = 1$ then $1^3 = 1$ and $3^1 = 3$ so true for $n = 1$

When $n = 2$ then $2^3 = 8$ and $3^2 = 9$ so true for $n = 1$

7) c)

Prove true for $n = 4$:

$4^3 = 64$ and $3^4 = 81$ and since $64 < 81$ true for $n = 4$

Assume true for $n = k$:

$k^3 < 3^k$

Prove true for $n = k + 1$:

$3k^3 < 3 \times 3^k$

$3k^3 < 3^{k+1}$

$k^3 + 2k^3 < 3^{k+1}$

But since $k > 3$ then $2k^3 > 6k^2$

$k^3 + 6k^2 < 3^{k+1}$

$k^3 + 3k^2 + 3k^2 < 3^{k+1}$

But also, since $k > 3$ then $3k^2 > 9k$

$k^3 + 3k^2 + 9k < 3^{k+1}$

$k^3 + 3k^2 + 3k + 6k < 3^{k+1}$

But also, since $k > 3$ then $6k > 18$

$k^3 + 3k^2 + 3k + 18 < 3^{k+1}$

$(k + 1)^3 + 17 < 3^{k+1}$

$(k + 1)^3 < 3^{k+1}$

which is $n^3 < 3^n$ with $n = k + 1$ substituted in.

True for $n = 1$ and $n = 2$. As the statement is true for $n = 4$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 4$

8)

Prove true for $n = 4$:

$4! = 24$ and $2^4 = 16$ since $24 > 16$

so true for $n = 4$

Assume true for $n = k$:

$k! > 2^k$

Prove true for $n = k + 1$:

$(k + 1)! = (k + 1)k!$

Since $k! > 2^k$

$(k + 1)k! > (k + 1)2^k$

Since $k > 1$ then $k + 1 > 2$ which is true so that

$(k + 1)2^k > 2 \times 2^k$

$(k + 1)! > 2^{k+1}$

which is $n! > 2^n$ with $n = k + 1$ substituted in

As the statement is true for $n = 4$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 4$

9)

Prove true for $n = 5$:

$5! = 120$ and $10 \times 5 = 50$

so true for $n = 5$

Assume true for $n = k$:

$k! > 10k$

Prove true for $n = k + 1$:

Since $k! > 10k$

$(k+1)k! > (k+1)10k$

$(k+1)! > 10k^2 + 10k$

Since $k \geq 5$ then $10k^2 \geq 50k$ so

$(k+1)! > 50k + 10k$

$(k+1)! > 10k + 50k$

Since $k \geq 5$ then $50k \geq 250$ so

$(k+1)! > 10k + 250$

$(k+1)! > 10k + 10$

$(k+1)! > 10(k+1)$

which is $n! > 10n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 5$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 5$

10)

Prove true for $n = 5$:

$3^5 = 243$ and $(5+1)^3 = 216$

so true for $n = 5$

Assume true for $n = k$:

$3^k > (k+1)^3$

Prove true for $n = k+1$:

$3^{k+1} = 3 \times 3^k > 3(k+1)^3$

$3^{k+1} > 3(k+1)^3$

$3^{k+1} > 3(k^3 + 3k^2 + 3k + 1)$

$3^{k+1} > 3k^3 + 9k^2 + 9k + 3$

$3^{k+1} > k^3 + 2k^3 + 9k^2 + 9k + 3$

But since $k > 4$ then $2k^3 > 8k^2$

$3^{k+1} > k^3 + 8k^2 + 9k^2 + 9k + 3$

$3^{k+1} > k^3 + 17k^2 + 9k + 3$

$3^{k+1} > k^3 + 3k^2 + 14k^2 + 9k + 3$

But since $k > 4$ then $14k^2 > 56k$

$3^{k+1} > k^3 + 3k^2 + 56k + 9k + 3$

$3^{k+1} > k^3 + 3k^2 + 65k + 3$

$3^{k+1} > k^3 + 3k^2 + 6k + 59k + 3$

But since $k > 4$ then $59k > 236$

$3^{k+1} > k^3 + 3k^2 + 6k + 236 + 3$

$3^{k+1} > k^3 + 3k^2 + 6k + 239$

$3^{k+1} > k^3 + 3k^2 + 6k + 8 + 231$

$3^{k+1} > (k+2)^3 + 231$

$3^{k+1} > (k+2)^3$

which is $3^n > (n+1)^n$ with $n = k+1$ substituted in.

As the statement is true for $n = 5$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 5$

11)

Prove true for $n = 4$:

$4! = 24$ and $4^2 + 4 = 20$

so true for $n = 4$

Assume true for $n = k$:

$k! > k^2 + k$

Prove true for $n = k + 1$:

Since $k! > k^2 + k$

$(k + 1)k! > (k + 1)(k^2 + k)$

$(k + 1)! > k^3 + 2k^2 + k$

Since $k \geq 4$ then $k^3 \geq 4k^2$ so

$(k + 1)! > 4k^2 + 2k^2 + k$

$(k + 1)! > k^2 + 5k^2 + k$

Since $k \geq 4$ then $5k^2 \geq 20k$ so

$(k + 1)! > k^2 + 20k + k$

$(k + 1)! > k^2 + 3k + 18k$

Since $k \geq 4$ then $18k \geq 72$ so

$(k + 1)! > k^2 + 3k + 72$

$(k + 1)! > k^2 + 3k + 2$

$(k + 1)! > k^2 + 2k + 1 + k + 1$

$(k + 1)! > (k + 1)^2 + (k + 1)$

which is $n! > n^2 + n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 4$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 4$

12)

Prove true for $n = 4$:

$\left(1 + \frac{1}{4}\right)^3 = \frac{125}{64} < 2$ so true $n = 4$

Assume true for $n = k$:

$2 > \left(1 + \frac{1}{k}\right)^3$

Prove true for $n = k + 1$:

Since $\frac{1}{k+1} < \frac{1}{k}$

$1 + \frac{1}{k} > 1 + \frac{1}{k+1}$

Since k is positive

$\left(1 + \frac{1}{k}\right)^3 > \left(1 + \frac{1}{k+1}\right)^3$

So

$2 > \left(1 + \frac{1}{k}\right)^3 > \left(1 + \frac{1}{k+1}\right)^3$

$\therefore 2 > \left(1 + \frac{1}{k+1}\right)^3$

which is $2 > \left(1 + \frac{1}{n}\right)^3$ with $n = k + 1$ substituted in.

As the statement is true for $n = 4$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 4$

13)

Prove true for $n = 2$:

$(1 + a)^2 = 1 + 2a + a^2 \Rightarrow (1 + a)^2 > 1 + 2a$

so true for $n = 2$

Assume true for $n = k$:

$(1 + a)^k > 1 + ak$

Prove true for $n = k + 1$:

$(1 + a)^{k+1} = (1 + a)(1 + a)^k$

Since $a > -1$, multiply the assumption through by $(1 + a)$

$(1 + a)(1 + a)^k > (1 + a)(1 + ak)$

$(1 + a)^{k+1} > 1 + a + ak + a^2k$

$(1 + a)^{k+1} > 1 + a(k + 1) + a^2k$

Since $a^2k \geq 0$

$(1 + a)^{k+1} > 1 + a(k + 1)$

which is $(1 + a)^n > 1 + an$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

14)

Prove true for $n = 2$:

$2! = 2$ and $\Rightarrow 2^2 = 4$

so true for $n = 2$

Assume true for $n = k$:

$k! < k^k$

Prove true for $n = k + 1$:

Since $k! < k^k$

$(k+1)k! < (k+1)k^k$

$(k+1)! < (k+1)k^k$

Since $(k+1)k^k < (k+1)(k+1)^k = (k+1)^{k+1}$

$(k+1)! < (k+1)^{k+1}$

which is $n! < n^n$ with $n = k + 1$ substituted in.

As the statement is true for $n = 2$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 2$

15)

Prove true for $n = 1$:

LHS: $\mathbf{ABA}^{-1}$ and RHS: $\mathbf{AB}^{1}\mathbf{A}^{-1}$ so true for $n = 1$

Assume true for $n = k$:

$(\mathbf{ABA}^{-1})^{k} = \mathbf{AB}^{k}\mathbf{A}^{-1}$

Prove true for $n = k + 1$:

$(\mathbf{ABA}^{-1})^{k+1} = (\mathbf{ABA}^{-1})^{k}(\mathbf{ABA}^{-1})$

$= \mathbf{AB}^{k}\mathbf{A}^{-1}(\mathbf{ABA}^{-1})$

$= \mathbf{AB}^{k}\mathbf{BA}^{-1}$

$= \mathbf{AB}^{k+1}\mathbf{A}^{-1}$ so true for $n = k + 1$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

16)

Prove true for $n = 1$:

LHS: $\mathbf{I} + \mathbf{A}$ and RHS: $\mathbf{I} + 1\mathbf{A}$ so true for $n = 1$

Assume true for $n = k$:

$\mathbf{I} + \mathbf{A} + \mathbf{A}^{2} + \cdots + \mathbf{A}^{k} = \mathbf{I} + k\mathbf{A}$

Prove true for $n = k + 1$:

$\mathbf{I} + \mathbf{A} + \mathbf{A}^{2} + \cdots + \mathbf{A}^{k} + \mathbf{A}^{k+1}$

$= \mathbf{I} + k\mathbf{A} + \mathbf{A}^{k+1}$

But since $\mathbf{A^{2}} = \mathbf{A}$ then $\mathbf{A^{3}} = \mathbf{A}$ so $\mathbf{A}^{k+1} = \mathbf{A}$

$\mathbf{I} + \mathbf{A} + \mathbf{A}^{2} + \cdots + \mathbf{A}^{k} + \mathbf{A}^{k+1} = \mathbf{I} + k\mathbf{A} + \mathbf{A}$

$= \mathbf{I} + (k + 1)\mathbf{A}$

Which is $\mathbf{I} + n\mathbf{A}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

17)

Prove true for $n = 1$:

LHS: $\det(\mathbf{A}_1) = \det(\mathbf{A})$

RHS: $(\det(\mathbf{A}))^1 = \det(\mathbf{A})$ so true for $n = 1$

Assume true for $n = k$:

$\det(\mathbf{A}_k) = (\det(\mathbf{A}))^k$

Prove true for $n = k + 1$:

$\det(\mathbf{A}_{k+1}) = \det(\mathbf{A} \times \mathbf{A}_k)$

$= \det(\mathbf{A}) \det(\mathbf{A}_k)$

$= \det(\mathbf{A}) (\det(\mathbf{A}))^k$

$= (\det(\mathbf{A}))^{k+1}$ so true for $n = k + 1$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

18)

Prove true for $n = 1$:

$\frac{\mathrm{d}}{\mathrm{d}x}(x^k) = kx^{k-1}$ and $\frac{k!}{(k-1)!}x^{k-1} = \frac{(k-1)!k}{(k-1)!}x^{k-1} = kx^{k-1}$

so true for $n = 1$

Assume true for $n = k$:

$\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^a) = \frac{a!}{(a-k)!}x^{a-k}$

Prove true for $n = k + 1$:

$\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}(x^a) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^a)\right)$

$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{a!}{(a-k)!}x^{a-k}\right)$

$= \frac{a!}{(a-k)!}(a-k)x^{a-k-1}$

$= \frac{a!}{(a-k)(a-k-1)!}(a-k)x^{a-k-1}$

$= \frac{a!}{(a-k-1)!}x^{a-k-1}$

$= \frac{a!}{(a-(k+1))!}x^{a-(k+1)}$

Which is $\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^a) = \frac{a!}{(a-n)!}x^{a-n}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

19)

Prove true for $n = 0$:

$\frac{0^2+0+2}{2} = 1$ which means there is an undivided plane

so true for $n = 0$

Assume true for $n = k$:

The plane is divided into $\frac{k^2+k+2}{2}$ regions.

Prove true for $n = k + 1$:

Since no two lines are parallel and no three meet at one point, the new line intersects all the existing k lines at distinct points.

These k lines divide the new line into $k + 1$ segments.

Each segment adds one new region to the plane:

$$\frac{k^2+k+2}{2} + (k+1) = \frac{k^2+k+2}{2} + \frac{2(k+1)}{2} = \frac{k^2+3k+4}{2} = \frac{k^2+2k+1+k+1+2}{2} = \frac{(k+1)^2+(k+1)+2}{2}$$

So this is true for $n = k + 1$.

As the statement is true for $n = 0$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 0$